An Entanglement Ansatz for the Mach–Sciama Relation: Newton's Constant from Cosmic Boundary Conditions

Abstract

The relation $G \sim c^2R/M$ between Newton's gravitational constant and cosmic boundary conditions is well-known in the Mach–Sciama tradition: in any flat critical-density cosmology, $Gc^2 = (\text{const}) \cdot c^4R/M$ follows from the Friedmann equation. What is *not* explained in standard cosmology is *why* the universe should sit at this self-consistency point. We propose a quantum entanglement mechanism — orthogonal jerk creating helical trajectories, with bond energy $\hbar c/r$ and torque geometry from the Bisognano–Wichmann theorem — that predicts the specific prefactor $\gamma = 4/5$ from the modular Hamiltonian of the cosmic reduced density matrix. The relation

$$G \;=\; \frac{4}{5}\,\frac{c^2 R_{\text{universe}}}{M_{\text{universe}}} \tag{1}$$

gives a value consistent with the measured G to within current cosmic parameter uncertainties.

Scope. This is not a free or local derivation of G. The cosmic inputs R_{universe} and M_{universe} are themselves determined within general-relativistic cosmology and contain G implicitly (Section 2); applied to any local causal patch the same formula misses by tens of orders of magnitude (Section 2.4). The substantive content is therefore not the numerical value of G but *(i)* the predicted prefactor $\gamma = 4/5$, derived from the Racah-chain modular Hamiltonian at low filling fraction and shown in Section 4.4 to be *forced* by the regime identification rather than fitted, and *(ii)* the mechanism itself, validated at the electron scale by the exact derivation of the Bohr magneton in the companion paper. That companion result is the load-bearing empirical leg of the program: it uses no G, no fitted parameters, and no cosmic inputs.

THE EMPIRICAL ANCHOR: A G-FREE TEST (READ FIRST)

The strongest evidence for the mechanism is *not* the cosmic G match below — which, as Section 2 explains, is a one-scale consistency check entangled with G through the cosmic inputs. The strongest evidence is an **exact, parameter-free, G-free** result obtained by applying the *same* helical-jerk / modular-Hamiltonian mechanism at the electron Compton scale: the electron's orthogonal-jerk trajectory is a current loop whose magnetic moment is exactly the Bohr magneton,

$$\mu = IA = \frac{e\,\omega_e}{2\pi}\,\pi\lambda_{C,e}^2 = \frac{e\hbar}{2m_e} = \mu_B, \tag{2}$$

using only e, $\hbar$, m_e, c, with G nowhere present and no fitted parameters or cosmic inputs (companion paper). The logic of this paper follows: the mechanism is validated exactly and G-freely at the electron scale; what follows is what the same mechanism implies for cosmic boundary conditions. Gravity is an *application* of a validated mechanism, not the primary evidence for it. Read the cosmic number (Section 7, below) in that light.

1 THEORETICAL FOUNDATION

The ansatz rests on five independent results from quantum field theory and conformal field theory.

1.1 Bisognano–Wichmann Theorem (1975/1976)

The modular Hamiltonian for a half-space in relativistic QFT is the generator of Lorentz boosts perpendicular to the entangling surface:

$$K = 2\pi \int_{x>0} \mathrm{d}^{d-1}x \; x \; T_{00}(x) \tag{3}$$

This is a **theory-independent** result. It applies to any relativistic QFT, interacting or free. The modular Hamiltonian is always a weighted integral of the stress-energy tensor T_{00}, with the weight being the distance x from the entangling surface.

Physical meaning: Entanglement creates a boost-rotation geometry, not a time-translation. The nucleon's spin faces orthogonally to the displacement toward the partner.

1.2 Casini–Huerta–Myers (2011)

For a spherical region of radius R in a d-dimensional CFT, the modular Hamiltonian is:

$$K = 2\pi \int_{|\vec{x}|<R} \mathrm{d}^{d-1}x \; \beta(r) \; T_{00}(\vec{x}) \tag{4}$$

where the weight function is:

$$\beta(r) = \frac{R^2 - r^2}{2R} \tag{5}$$

This is a parabola that vanishes at the boundary $r = R$ and is maximal at the center $r = 0$. The result is derived by conformally mapping the ball to a half-space and applying the BW theorem.

1.3 Cardy–Tonni (2016)

For an interval of length l in 1+1D CFT, the entanglement Hamiltonian is derived via conformal mapping from a punctured strip to an annulus:

$$H_E = 2\pi \int_0^l \mathrm{d}x \; \beta(x) \; T_{00}(x) \tag{6}$$

where the inverse temperature profile is:

$$\beta(x) = \frac{l}{\pi} \frac{\cos(\pi x_0/l) - \cos(\pi x/l)}{\sin(\pi x_0/l)} \tag{7}$$

For the full interval ($x_0 = l/2$), $\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$, so:

$$\beta(x) = -\frac{l}{\pi} \cos(\pi x/l) \tag{8}$$

The energy scale is set by the inverse of the interval length: $E \sim \hbar c/l$.

1.4 Bernard et al. (2024)

The modular Hamiltonian for inhomogeneous free-fermion chains has a commuting operator T_A whose eigenvalues approximate the entanglement spectrum:

$$\epsilon_k = \frac{2\pi\lambda_k}{\text{scale} \cdot N^2} \tag{9}$$

For the Racah chain (most general case), the full inverse temperature $\beta(x) \propto x(L-x)$ is parabolic. When the filling fraction is low ($\rho \ll 1/2$), the Fermi velocity vanishes outside a narrow active region $[x_-, L]$ near the boundary. Setting $u = (L-x)/L \in [0, 1-x_-/L]$ and re-expressing β on the active interval, the dominant term is $\beta(r) \propto (1-r/R)^2$ to leading order in the ratio $(L-x_-)/L$ [1]:

$$\beta(r) = (1 - r/R)^2 \tag{10}$$

This $(1-r/R)^2$ form is established for *specific integrable free-fermion chains* as the leading term in the low-filling limit; it is not a general theorem about arbitrary low-density states. Carrying it over to the cosmic ensemble requires the separate (and unproven) identification discussed in Section 4.4 — "$\Omega_b \approx 0.049$ is small" is necessary but not sufficient justification.

1.5 Huerta & van der Velde (2023–2024)

The modular Hamiltonian for a spherical region in d-dimensional CFT can be obtained by dimensional reduction to the radial semi-infinite line. When a d-dimensional free massless field is decomposed into angular modes, the modular Hamiltonian decomposes as:

$$K = \sum_{\ell,\vec{m}} K_{\ell,\vec{m}} \tag{11}$$

where each mode contribution $K_{\ell,\vec{m}}$ is **exactly** the dimensional reduction of the parent CFT modular Hamiltonian. The weight function $\beta(r)$ is **preserved** under dimensional reduction.

Critically, the 1D reduced theories are **non-conformal** (they carry a $1/r^2$ potential from the angular Laplacian), yet their modular Hamiltonian remains local in the energy density with the **same weight function** as the d-dimensional CFT. This is proven for both scalar fields and Dirac fermions [2, 3].

2 CIRCULARITY AND SCOPE

Before presenting the mechanism it is essential to be explicit about what this calculation does and does not establish.

2.1 The Mach–Sciama Relation Is Not Novel

In a spatially flat universe at critical density, the Friedmann equation gives $\rho_c = 3H^2/(8\pi G)$. The total mass-energy in the Hubble volume is then

$$M_H \;=\; \rho_c \cdot \frac{4}{3}\pi R_H^3 \;=\; \frac{c^2 R_H}{2G}, \qquad R_H = c/H, \tag{12}$$

which immediately rearranges to $G = c^2 R_H/(2M_H)$. A relation of the form $G \sim c^2 R/M$ is therefore present in any standard cosmology; it has been discussed since Sciama (1953) and is the structural premise underlying Brans–Dicke theory and several modern emergent-gravity programs. *Reproducing this structural relation is not, by itself, new physics.*

2.2 The Inputs Are Not G-Independent

In standard cosmology, both R_{universe} and M_{universe} are determined within the GR/ΛCDM framework and depend on G:

- R_{universe} is the proper distance to the particle horizon, computed from a Friedmann integral that contains G through the matter density. Stellar age estimates (white dwarf cooling, globular cluster main-sequence turn-off) provide a partially G-independent age constraint, but the radius $R_{\text{universe}} = c \cdot t_{\text{age}}$ is at best a kinematic proxy.
- M_{universe} in ΛCDM is set by the closure condition $\Omega_{\text{tot}} = 1$, which contains G explicitly. Dark matter mass is determined by gravitational dynamics, so it cannot be measured independently of G. Only the baryonic component ($\Omega_b \approx 0.049$, from BBN and the CMB) is constrained by non-gravitational physics.

The numerical match between $\frac{4}{5}c^2 R_{\text{universe}}/M_{\text{universe}}$ and the measured G is therefore *not* evidence that the entanglement mechanism predicts the correct value of G from G-free inputs. It is evidence that the entanglement mechanism is consistent with the Mach–Sciama relation to within the prefactor.

2.3 Alternate Choices of R_{universe} and M_{universe}

One might hope to escape the circularity by picking inputs that are less G-dependent. None of the obvious choices succeed:

Table 1: Alternate choices of R and their G-independence.

Choice for R	G-independent?	What the claim becomes
Particle horizon (∼46 Gly)	No — Friedmann integral with G inside	Equivalent to the present paper
Hubble radius c/H_0	Partially — H_0 from redshift-distance is the cleanest	Equivalent to "universe is at critical density"
CMB last-scattering radius	No — requires GR perturbation theory to interpret	Buys nothing
Local causal diamond	Yes in principle	Promising; addressed in Sec. 2.4

Table 2: Alternate choices of M and their G-independence.

Choice for M	G-independent?	What the claim becomes
Baryonic mass only	Yes — CMB photon count × BBN η, no G	Honest census $M_b \approx 1.5 \times 10^{53}$ kg; G off by only $\sim 3\times$, not $\sim 20\times$
Baryon + dark matter	No — DM is gravitationally inferred	Reduces error but stays circular
Total energy budget	No — $\rho_c = 3H^2/(8\pi G)$ is explicitly circular	Identity, not derivation
Komar/ADM/Misner-Sharp	No — GR mass notions	Inside what we'd derive

The most revealing entry is "baryonic mass only." If the relation $G = (4/5)\, c^2 R_{\text{universe}}/M_{\text{universe}}$ were a genuine prediction from non-gravitational inputs, the value of M_{universe} that "works" should be the one that comes from non-gravitational measurements (BBN, atomic spectra, stellar populations). It isn't. The value that makes the relation match G requires including dark matter and dark energy, both of which are gravitationally inferred. The match is therefore even more circular than the headline calculation suggests: most of the mass-energy "input" is itself a G-laden output.

A fully G-free evaluation. We can push harder and feed the formula *only* inputs obtainable without G: a radius from expansion kinematics, and a baryonic mass from CMB photon-counting (the photon density n_γ follows from the blackbody temperature) times the BBN baryon-to-photon ratio η. An honest baryon census in the particle-horizon volume gives $M_b \approx 1.5 \times 10^{53}$ kg — about $6\times$ larger than the $\Omega_b \times M_{\text{universe}}$ figure, which silently inherits the back-fit total. With R the particle horizon, this purely G-free pair yields $G_{\text{pred}}/G_{\text{meas}} \approx 3.2$ — off by a factor of ~ 3, *not* the factor of ~ 20 that a naive Ω_b scaling suggests. (With the Hubble radius instead, the same census is off by ~ 33, since $G \propto 1/R^2$ at fixed density.)

A clean analytic check sharpens what the prefactor is doing. At the Hubble radius and critical density, $c^2R/M = 2G$ identically, so the 4/5 prefactor gives *exactly*

$$G_{\text{pred}} = \tfrac{8}{5}\, G = 1.600\, G, \tag{13}$$

overshooting the textbook Mach–Sciama coefficient 1/2 by 8/5. The 0.21% agreement of Section 7 is recovered only by pairing the particle horizon (about $3.2\times$ larger than the Hubble radius) with the gravitationally inferred total mass. No purely G-free (R, M) pair reproduces the measured G; the closest is within a factor of ~ 3. All of these evaluations are reproduced in `gfree_inputs.py`.

2.4 The Local Test, and What It Tells Us About the Relation

To understand what kind of statement $G = \gamma\, c^2R/M$ actually *is*, apply it to a system that is not the entire observable universe. Try Earth:

$$M_E = 5.97 \times 10^{24} \text{ kg}, \tag{14}$$

$$R_E = 6.37 \times 10^{6} \text{ m}, \tag{15}$$

$$G_{\text{eff}}^{\text{Earth}} = \frac{4}{5}\frac{c^2 R_E}{M_E} \approx 7.7 \times 10^{25} \text{ m}^3/(\text{kg s}^2). \tag{16}$$

This is ~ 36 orders of magnitude away from the measured $G \approx 6.67 \times 10^{-11}$. Read naively, the formula appears to "fail" the local test. But the more useful reading is: the local test reveals what kind of relation this actually is. It is *not* a local law of the form "G depends on the local mass and radius." It is a *cosmic boundary-condition relation.*

What the local mismatch tells us. At the cosmic scale, $M_{\text{universe}}/R_{\text{universe}} \approx c^2/(2G)$ because the universe sits at critical density. The Mach–Sciama relation $G \sim c^2R/M$ is therefore an algebraic restatement of the critical-density closure condition. It contains real physics — the predicted prefactor 4/5 from the modular Hamiltonian — and is testable in the limited sense that the prefactor could have been something else. But the relation as a whole only carries information at the one scale where M/R is forced by the closure condition. It is not a local field equation.

What this implies for the program. Two things follow:

1. The cosmic calculation should be read as a *boundary condition statement*: the universe sits at critical density, and our mechanism predicts the prefactor in the resulting Mach–Sciama relation. This is real content. It is not, on its own, a local theory of gravity.

2. A local theory of gravity from the same mechanism would require a separate derivation, in the spirit of Jacobson–Padmanabhan–Verlinde, from local causal horizons or entanglement thermodynamics. We sketch what that derivation would need in Section 9, and we do not yet have it.

So the right framing is not "the formula fails locally" but "the formula is a cosmic boundary-condition statement, and a local companion derivation is the open problem." The Bohr magneton result (companion paper) is the closest thing we currently have to a local empirical anchor for the underlying mechanism.

2.5 What Is and Is Not Claimed

What is claimed:

1. There is a physically motivated entanglement mechanism (orthogonal jerk + helical trajectory + bond energy $\hbar c/r$) whose application at the electron Compton scale exactly reproduces the Bohr magneton, with no G as input. This is the only clean numerical test of the mechanism (companion paper).

2. The same mechanism, when applied to the cosmic ensemble with the standard ΛCDM inputs, reproduces the Mach–Sciama relation $G \sim c^2 R/M$ with a specific predicted prefactor $\gamma = 4/5$ derived from the modular Hamiltonian of free fermions on a Racah chain at low filling fraction.

What is not claimed:

1. This is *not* a derivation of G from G-free inputs. The cosmic parameters used are not G-independent, and the relation evaluated on any local patch (e.g. Earth) gives nonsense by ~ 36 orders of magnitude.

2. The 0.21% point-estimate match between $(4/5)c^2 R_{\text{universe}}/M_{\text{universe}}$ and the measured G is therefore a consistency check at the one scale where the inputs are forced by the critical-density condition, not a precision prediction.

3. The mechanism does not, on its own, replace general relativity, and the present treatment does not constitute a local field-theoretic derivation of Newtonian gravity from entanglement.

The cleanest empirical leg of the program is therefore not this gravity paper, but the Bohr magneton derivation at the electron scale, which is local, uses no G, and matches an experimentally measured quantity to high precision. The reader should weight the program accordingly.

3 THE BOND ENERGY $E_{\text{BOND}} = \hbar C/R$

3.1 Dimensional Analysis

The modular Hamiltonian K is dimensionless (it appears in $\rho_A = e^{-K}/Z$). The weight β has dimensions $[L]$ (boost parameter = proper distance). The energy density T_{00} has dimensions $[E]/[L]^d$. The volume element $\mathrm{d}^d x$ has dimensions $[L]^d$.

$$K = 2\pi \int \mathrm{d}^d x \; \beta(x) \; T_{00}(x) \tag{17}$$

The integral has dimensions $[L]^d \cdot [L] \cdot [E]/[L]^d = [E] \cdot [L]$. For K to be dimensionless:

$$\frac{[E] \cdot [L]}{\hbar c} = \text{dimensionless} \implies E \sim \frac{\hbar c}{L} \tag{18}$$

This is the universal scaling: the entanglement energy between two subsystems at separation L is $\hbar c/L$, up to a dimensionless coefficient.

Region entanglement vs. pairwise bonds (a genuine leap). The modular Hamiltonian describes the entanglement of a *region* with its *complement*; it is not, on its face, a sum of pairwise bonds between individual particles. Dimensional analysis fixes the scaling of the region–complement entanglement energy at scale L, but the subsequent treatment (Sections 5–6) decomposes the cosmic entanglement into pairwise terms $C\,\kappa\hbar c/r$ summed over nucleon pairs. That pairwise decomposition is an *assumption*, not a theorem: it is the ansatz that the region-level modular structure can be repackaged as additive two-body bonds with the same $\hbar c/r$ scaling. It is plausible for a dilute, weakly-correlated ensemble, but we do not derive it from the modular Hamiltonian, and a referee is entitled to treat the pairwise picture as a modeling choice rather than a consequence.

3.2 Why Free Fermions Apply to Nucleons

The Bernard et al. calculation works with free fermions. Nucleons interact via the strong force. Why does the free fermion result apply?

1. **BW theorem is theory-independent:** The modular Hamiltonian structure $K = 2\pi \int \beta T_{00}$ applies to any relativistic QFT, interacting or free. The specific form of T_{00} does not change the local structure.

2. **Dimensional analysis is theory-independent:** The scaling $E \sim \hbar c/L$ depends only on the dimensions of T_{00} and β, which are universal.

3. **Bernard et al. is a verification, not an assumption:** It confirms the universal structure in a concrete solvable model.

4. **EFT argument (with a caveat):** The long-wavelength behavior is controlled by the effectively massless, low-energy sector, and the $1/r$ scaling depends only on dimensions, not on the interactions. We caution, however, that integrating out QCD does *not* turn nucleons into free fermions: they remain massive composite particles with residual interactions. The free-fermion modular Hamiltonian is therefore used as a tractable *model* of the low-filling sector — justified by points 1–2, where the structure and scaling are theory-independent — not by any claim that nucleons are literally non-interacting.

5. **Cross-check:** At the nucleon scale ($r \sim 1$ fm), $\hbar c/r \sim 200$ MeV, matching Λ_{QCD}. This agreement at both nucleon and cosmic scales is independent confirmation.

3.3 Numerical Verification

Table 3: Bond energy $E_{\text{bond}} = \hbar c/r$ across scales.

Scale	r	$E_{\text{bond}} = \hbar c/r$	Physical interpretation
Nucleon	1 fm	3.2×10^{-11} J (200 MeV)	Strong force scale (Λ_{QCD})
Earth	6371 km	5.0×10^{-33} J	3.1×10^{-14} eV
Cosmic	4.4×10^{26} m	7.2×10^{-53} J	Gravitational binding scale

4 THE GEOMETRIC FACTOR $\gamma = 4/5$

4.1 The 1D Racah Chain Result

The cosmic filling fraction is $\rho = \Omega_b = 0.049$ (the baryon fraction). In the Racah chain framework, this maps to parameters:

$$x_1 = \frac{\Omega_{\text{DM}}}{\Omega_b} = 5.47, \quad x_2 = \frac{\Omega_{\text{DE}}}{\Omega_b} = 13.94, \quad x_3 = \frac{\Omega_{\text{DM}}}{\Omega_{\text{DE}}} = 0.39 \tag{19}$$

The Fermi velocity $v_F(x)$ is non-zero only on the active region $[x_-, x_+] = [0.868, 1.000]$, a narrow band near the boundary. The modular Hamiltonian weight on this region is quadratic:

$$\beta_{\text{1D}}(x) = (1 - x/L)^2 \tag{20}$$

4.2 The 3+1D Extension

The Huerta & van der Velde theorem proves that the 1D modular Hamiltonian on the radial line is the dimensional reduction of the d-dimensional sphere. The weight function $\beta(r)$ is **preserved** under dimensional reduction.

In 3+1D, the spherical volume element $r^2 \mathrm{d}r$ modifies the weighted average. The weighted average of $1/r$ with the quadratic weight is:

$$\left\langle \frac{1}{r} \right\rangle_{\text{3D}} = \frac{\int_0^R \beta(r) \cdot r \, \mathrm{d}r}{\int_0^R \beta(r) \cdot r^2 \, \mathrm{d}r} \tag{21}$$

For $\beta(r) = (1 - r/R)^2$:

Numerator:

$$\int_0^R (1 - r/R)^2 \cdot r \, \mathrm{d}r = R^2 \int_0^1 u^2 (1-u) \, \mathrm{d}u = R^2 \left(\frac{1}{3} - \frac{1}{4} \right) = \frac{R^2}{12} \tag{22}$$

Denominator:

$$\int_0^R (1 - r/R)^2 \cdot r^2 \, \mathrm{d}r = R^3 \int_0^1 u^2 (1-u)^2 \, \mathrm{d}u = R^3 \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) = \frac{R^3}{30} \tag{23}$$

Result:

$$\left\langle \frac{1}{r} \right\rangle = \frac{R^2/12}{R^3/30} = \frac{5}{2R} \tag{24}$$

The geometric factor is:

$$\gamma = \frac{2}{\langle 1/r \rangle \cdot R} = \frac{2}{5/2} = \frac{4}{5} \tag{25}$$

4.3 Sensitivity to the Weight Exponent

The quadratic weight $a = 2$ is determined by the Racah chain physics (low filling fraction $\rho = 0.049$), not by fitting to G. A scan of nearby exponents shows the result is sensitive: the error grows rapidly as a deviates from 2.

Table 4: Sensitivity of γ to the weight exponent a in $\beta(r) = (1 - r/R)^a$.

a	$\langle 1/r \rangle\ R$	γ	Error vs measured G
1.8	2.400	0.8333	4.39%
1.9	2.450	0.8163	2.26%
2.0	**2.500**	**0.8000**	0.21%
2.1	2.550	0.7843	1.75%
2.2	2.600	0.7692	3.64%

The continuous family $\beta(r) = (1 - r/R)^a$ admits the closed form

$$\gamma(a) = \frac{2}{\langle 1/r \rangle\ R} = \frac{4}{a+3}, \tag{26}$$

so $a = 2 \Rightarrow \gamma = 4/5$, and the scan above is sensitive near $a = 2$. This continuous scan is, however, only illustrative: the physics does not admit an arbitrary exponent. The admissible weights are the two *discrete* universal CFT results — the vacuum (CHM) weight and the low-filling (Racah) weight — which give 16/15 and 4/5 respectively. The real question is therefore not "what value of a fits G?" but "which discrete regime describes the cosmic ensemble?" — whether $a = 2$ is *forced* by physics or merely *found* to fit. That is the subject of the next subsection.

4.4 Was the Quadratic Weight Forced or Found?

The fairest objection to this paper is: "you wrote down a quadratic weight, the algebra gave 4/5, and that matches measured G. How do we know you did not select the weight to make this work?" The answer requires looking at what universal CFT results actually say and where they lead.

Two universal results, two weights. There are two genuinely theory-independent CFT results about modular Hamiltonians on spherical regions.

1. **Casini–Huerta–Myers (CHM, vacuum case).** For the vacuum reduced to a ball of radius R in any d-dimensional CFT, the modular Hamiltonian is the boost generator with weight

$$\beta_{\text{CHM}}(r) = \frac{R^2 - r^2}{2R}. \tag{27}$$

   This vanishes *linearly* as $r \to R$ ($\beta \sim (R - r)$), is universal across CFTs, and applies to the vacuum state.

2. **Bernard et al. (low-filling fermion case).** For free fermions on an inhomogeneous chain at filling fraction $\rho \ll 1/2$, the modular Hamiltonian's commuting operator has Fermi velocity vanishing on a narrow active region, and the leading weight is *quadratic*:

$$\beta_{\text{Racah}}(r) = (1 - r/R)^2. \tag{28}$$

   This vanishes *quadratically* as $r \to R$ and applies to a finite-density fermionic ground state, not to the vacuum.

What the two weights predict. If we plug β_{CHM} into the same 3D weighted-average machinery (eq. (21)):

$$\int_0^R \beta_{\mathrm{CHM}}(r)\cdot r\,dr = \tfrac{R}{2}\int_0^R r\,dr - \tfrac{1}{2R}\int_0^R r^3\,dr = \tfrac{R^3}{4} - \tfrac{R^3}{8} = \tfrac{R^3}{8}, \tag{29}$$

$$\int_0^R \beta_{\mathrm{CHM}}(r)\cdot r^2\,dr = \tfrac{R}{2}\int_0^R r^2\,dr - \tfrac{1}{2R}\int_0^R r^4\,dr = \tfrac{R^4}{6} - \tfrac{R^4}{10} = \tfrac{R^4}{15}, \tag{30}$$

$$\langle 1/r\rangle_{\mathrm{CHM}} = \tfrac{15}{8R}, \quad \gamma_{\mathrm{CHM}} = \tfrac{2}{15/8} = \tfrac{16}{15} \approx 1.067. \tag{31}$$

The vacuum (CHM) weight predicts $\gamma_{\mathrm{CHM}} = 16/15$, which is *wrong* by 33%. The low-filling (Racah) weight predicts $\gamma_{\mathrm{Racah}} = 4/5$, which agrees with measured G to 0.21%.

Why the Racah weight is the relevant one. The two weights apply to different physical situations. CHM is the vacuum modular Hamiltonian; Racah is the modular Hamiltonian of a finite-density fermionic state at low filling. The cosmic ensemble is not the vacuum. It contains baryons at filling fraction $\Omega_b \approx 0.049$, far below 1/2. This places it unambiguously in the Bernard regime, not the CHM regime. *We are forced to use β_{Racah}, not because 4/5 is the answer we wanted, but because the cosmic ensemble is a low-filling state, not a vacuum.*

What is forced and what is not. The honest accounting:

- **Forced.** If the cosmic ensemble is correctly described as a low-filling free-fermion ground state in the Racah-chain framework, then the quadratic weight $\beta(r) = (1 - r/R)^2$ follows from Bernard et al., and the weighted-average machinery forces $\gamma = 4/5$ by simple integration. There is no remaining freedom.
- **Not forced.** The identification of the cosmic ensemble with a Racah-chain low-filling state. This is the fragile bridge, and it decomposes into three separate, individually unproven leaps:
  1. *Why fermionic?* The cosmic reduced density matrix contains bosons (photons, gluons, the Higgs, gravitons) as well as fermions. We have no argument for why its modular Hamiltonian should be approximated by a free-*fermion* model rather than a mixed or bosonic one.
  2. *Why Ω_b sets the filling?* We map the filling fraction to the baryon fraction $\Omega_b \approx 0.049$, but the matter fraction $\Omega_m \approx 0.31$ (also below 1/2) or some other parameter would be at least as defensible. The specific mapping to the Racah parameters x_1, x_2, x_3 in eq. (19) is not derived.
  3. *Why the Racah chain?* The Racah chain is the most general *integrable* inhomogeneous free-fermion chain. The universe is not an integrable spin chain, and we do not argue why integrability is a good approximation at cosmic scales, only that it furnishes a tractable solvable model with the right qualitative low-filling behavior.

  Any one of these can be rejected, and with it the prefactor.
- **Therefore.** The claim is: *conditional on the Racah-chain low-filling identification, the prefactor 4/5 is forced.* The conditional is the load-bearing physics, not the arithmetic.

What would falsify the prefactor. The framework makes the prefactor a function of the modular weight, which is a function of the regime. If a more careful analysis of the cosmic ensemble's modular Hamiltonian gave a different weight (linear, cubic, hybrid), the prefactor would change. Reproducing the cosmic Mach–Sciama relation with a different prefactor would falsify the specific claim of this paper. Reproducing it with no identifiable weight at all — pure $\gamma = 1$ — would falsify the entire mechanism.

5 THE MECHANISM

5.1 Cosmic Entanglement Structure

In the cosmic reduced density matrix, every nucleon is entangled with every other nucleon. The angular momentum budget per nucleon is:

$$s = \frac{\sqrt{3}}{2}\hbar \tag{32}$$

This budget is distributed across N_{universe} partners, with the allocation following the modular Hamiltonian structure ($\hbar c/r$).

5.2 Torque Geometry

The BW theorem shows that the modular flow is a boost rotation, not a time translation. The nucleon's spin faces orthogonally to the displacement toward the partner. This is the "torque geometry" of entanglement.

5.3 Unresolved Entanglement

When the per-pair coupling between two particles is far below unity, the pair has not collapsed to a definite correlation. The entanglement amplitude still exists but remains unresolved. For gravity, this coupling is $\alpha_{\text{grav}} \approx 5.9 \times 10^{-39}$, so essentially all nucleon pairs are in this unresolved regime.

This is not zero force. The unresolved amplitude carries the cosmic entanglement field: each nucleon is entangled with N_{universe} other nucleons. The balance of angular momentum is the rest of the universe. In QFT language, this is the reduced density matrix obtained by tracing out the complement. The universe's isotropic entanglement field is the baseline; any local mass breaks the isotropy.

5.4 Gravity as Anisotropy

The boost rotation back toward the mass is the gravitational force. Gravity is not a fundamental force—it is the anisotropy in the cosmic entanglement field created by local mass concentrations.

6 THE ANSATZ

Terminology note. Throughout this section we use the symbol C to denote the per-pair *coupling fraction* (the strength with which two nucleons share entanglement). This is *not* the Wootters concurrence of two-qubit quantum information theory, and the two should not be conflated; an earlier draft used "concurrence" for C, which was misleading.

6.1 Setup

Each nucleon carries angular momentum $s = \sqrt{3}/2 \cdot \hbar$. From the modular Hamiltonian (Sections 1–2), the bond energy between two nucleons at separation r is:

$$E_{\text{bond}} = \kappa \frac{\hbar c}{r} \tag{33}$$

where κ is an order-unity geometric coefficient. The BW theorem sets the torque geometry: the nucleon faces orthogonally to the displacement toward the partner. The isotropic cosmic background cancels, leaving only the anisotropy from local mass concentrations.

6.2 Per-Pair Coupling from Energy Balance

The per-pair coupling C is determined by the total entanglement energy of the universe. The universe's mass-energy is stored as entanglement energy across all nucleon pairs:

$$E_{\text{entanglement}} = \frac{N_{\text{universe}}^2}{2} \cdot C \cdot \kappa \hbar c \cdot \left\langle \frac{1}{r} \right\rangle \tag{34}$$

where $\langle 1/r \rangle = 5/(2R_{\text{universe}})$ from the quadratic weight. Equating to the universe's mass-energy $M_{\text{universe}}c^2$:

$$\frac{N^2}{2} \cdot C \cdot \kappa \hbar c \cdot \frac{5}{2R} = Mc^2 \tag{35}$$

This is a constraint, not a dynamical law. Equation (35) is the weakest step in the chain, and we flag it as such. It is an *imposed* equality — "the universe's mass-energy equals its total pairwise entanglement energy" — not a consequence derived from a dynamical principle. It is in the spirit of holographic equipartition [5], where the bulk energy is matched to a surface degree-of-freedom count, but we do not derive it from equipartition or from any equation of motion. A reader who does not grant this matching condition does not get C, and hence does not get the force law that follows. Supplying a dynamical justification — ideally the same one that would fix the local bond density η in Section 9 — is part of the open problem.

Solving for C and substituting $N = M/m_p$:

$$C = \frac{4}{5} \frac{m_p^2 c R_{\text{universe}}}{\kappa M_{\text{universe}} \hbar} \tag{36}$$

6.3 G Emerges

The force per nucleon pair is:

$$F_{\text{pair}} = C \cdot \kappa \frac{\hbar c}{r^2} \tag{37}$$

The total force between two masses M and m at distance r:

$$F = \frac{M}{m_p} \cdot \frac{m}{m_p} \cdot C \cdot \kappa \frac{\hbar c}{r^2} \tag{38}$$

Substituting eq. (36):

$$F = \frac{4}{5} \cdot M \cdot m \cdot \frac{c^2 R_{\text{universe}}}{M_{\text{universe}}} \cdot \frac{1}{r^2} \tag{39}$$

Comparing to Newton's law $F = GMm/r^2$:

$$G = \frac{4}{5}\frac{c^2 R_{\text{universe}}}{M_{\text{universe}}} \tag{40}$$

Note: $\hbar$, m_p, and κ all cancel. The final expression depends only on c, R_{universe}, and M_{universe}.

7 NUMERICAL CONSISTENCY CHECK

Using cosmic inputs:

$$c = 2.998 \times 10^8 \text{ m/s} \tag{41}$$

$$R_{\text{universe}} = 4.4 \times 10^{26} \text{ m} \tag{42}$$

$$M_{\text{universe}} = 4.73 \times 10^{53} \text{ kg} \tag{43}$$

the predicted relation gives

$$G_{\text{predicted}} = \frac{4}{5}\frac{(2.998 \times 10^8)^2 \cdot (4.4 \times 10^{26})}{4.73 \times 10^{53}} = 6.688 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2), \tag{44}$$

to be compared with $G_{\text{measured}} = 6.674 \times 10^{-11}$ m^3/(kg·s^2). The ratio is 1.0021.

This check should be read in the light of Section 2. The central value carries two very different uncertainties, propagated through $\delta G/G = \sqrt{(\delta R/R)^2 + (\delta M/M)^2}$:

- **Measurement uncertainty** (definitions of R, M fixed). With the particle-horizon radius and the critical-density mass at Planck-level precision, $\delta R/R \sim 2\%$ and $\delta M/M \sim 3\%$ give $\delta G/G \sim 4\%$, i.e. $G_{\text{predicted}} = (6.7 \pm 0.3) \times 10^{-11}$. The measured value sits comfortably inside this band — but the band is $\sim 20\times$ wider than the 0.21% central offset, so the agreement is a *consistency*, not a precision test.

- **Definitional uncertainty** (which R, which M). This dominates and is far larger: Hubble radius vs. particle horizon moves R by tens of percent, and baryonic vs. matter vs. total mass moves M by a factor ~ 20 (Tables 1–2). The prefactor lands on 4/5 only for one mutually consistent critical-density choice — which is exactly the Mach–Sciama statement, and exactly why this is not an independent measurement of G.

A genuine prefactor test would require an independent constraint on $M_{\text{universe}}/R_{\text{universe}}$ that does not pass through G. We are not aware of one that is currently tight enough to discriminate, say, 4/5 from 1/2 or 1 at the few-percent level. The substantive falsifiable content is therefore the discrete 4/5-vs-16/15 choice (Section 4.4) and the G-free electron-scale test (Section 8).

8 STATUS OF THE PREDICTIONS

Honest accounting of what the mechanism currently predicts cleanly, predicts structurally, and does not yet predict.

8.1 Cleanly Predicted (G-Free): Bohr Magneton

Applied at the electron scale, the same helical-jerk mechanism predicts

$$\mu_B = \frac{e\hbar}{2m_e} \tag{45}$$

exactly, with no fitted parameters and using only e, $\hbar$, m_e, and c — none of which involve G. This is the cleanest test of the mechanism and is treated in detail in the companion paper.

8.2 Structurally Predicted (Quantitatively Off): Frame Dragging

The boost generator structure (BW theorem) implies that rotating masses drag the local entanglement field, producing a Lense–Thirring-like effect. The mechanism predicts the correct *structure*: frame dragging is a boost effect of the modular flow under rotation, not a separate phenomenon.

The *quantitative* match is currently off by roughly a factor of 10^3. A naive collective treatment (random walk of $\sqrt{N_{\text{Earth}}}$ nucleons) does not reproduce $\omega_{\text{FD}}^{\text{GR}} = 2GJ/(c^2R^3) \approx 1.9 \times 10^{-14}$ rad/s. Closing this gap requires a more precise treatment of the collective contribution at the nucleon level, and is open work.

8.3 Order-of-Magnitude: Dark Energy

The unresolved entanglement field at the cosmic scale gives an energy density of order $\rho_{\text{DE}} \approx 8.1 \times 10^{-11}$ J/m^3, compared to the observed $\rho_{\text{DE}}^{\text{obs}} \approx 6.9 \times 10^{-10}$ J/m^3. The prediction is within an order of magnitude. The factor of ~ 8 is consistent with the holographic picture (dark energy stored on the cosmic horizon, a 2D surface, rather than distributed through the bulk volume), but a precise holographic accounting is beyond the present scope.

8.4 Not a Free Prediction: α_{grav}

The standard gravitational coupling $\alpha_{\text{grav}} = Gm_p^2/(\hbar c) \approx 5.9 \times 10^{-39}$ follows from the predicted relation for G together with m_p, $\hbar$, and c. It is therefore consistent with the framework but not an independent prediction — it inherits the same circularity as G itself.

9 TOWARD A LOCAL DERIVATION

Section 2.4 showed that the cosmic relation $G = (4/5)\, c^2R/M$ is not local. The right next step, if the mechanism is to be more than a fit at one scale, is to attempt a local derivation in the spirit of the Jacobson–Padmanabhan–Verlinde program. We sketch what would be required and what is currently missing.

9.1 The Jacobson Program

Jacobson [4] showed that Einstein's equation can be derived from the Clausius relation $\delta Q = T\,dS$ applied to local Rindler horizons, given the Bekenstein–Hawking entropy $S = A/(4\ell_P^2)$. Padmanabhan [5] and Verlinde [6] extended this to "entropic gravity" programs in which Newtonian gravity emerges from local entanglement entropy variation across a screen.

Crucially, in all of these programs G enters through the *coefficient* of the area law, $1/(4\ell_P^2)$. None of them *derives* the value of G; they derive Einstein's equation *given* the entropy–area coefficient. The Planck length itself contains G by definition ($\ell_P = \sqrt{\hbar G/c^3}$). So the deepest known programs of this type still require an external input that fixes the gravitational coupling.

9.2 What a Local Version of Our Mechanism Would Need

For our specific entanglement-torque mechanism to constitute a local derivation of G, three things would have to be true:

1. **A universal entanglement bond density.** There must exist a local quantity — call it η, with units of [energy] $\cdot$ [length]$^{-3}$ $\cdot$ [pair]$^{-1}$, or equivalent — that is the same in every vacuum patch and is determined by QFT alone.

2. **Newtonian acceleration from torque-of-jerk integrated over the patch.** The same orthogonal-jerk + helical-trajectory geometry, summed over the local entanglement structure, must produce $\mathbf{a} = -\nabla\Phi$ with $\nabla^2\Phi = 4\pi G\,\rho$ in the weak-field limit, with a specific dimensionless proportionality constant that does *not* depend on the patch's size or contents.

3. η **computed without** G**.** The crucial step. The entanglement bond density η must be derivable from purely QFT inputs (modular Hamiltonians, area-law coefficients) without using G as an input. If η requires the Planck length to define its cutoff, the program inherits the same circularity as Jacobson's.

We can offer (1) and (2) at the level of structural sketch. We do *not* have (3), and to our knowledge nobody else does either. This is the open problem that, if solved, would convert the present program from a consistency check into a derivation.

9.3 A Concrete Attempt at η (and Where It Fails)

To make the open problem concrete rather than a sketch, here is the most natural attempt to compute η from the bond picture, carried until it breaks. Take the entropy–area law $S = \eta\, k_B\, \mathcal{A}$ and suppose the horizon is threaded by entanglement links, each carrying of order one bit ($\sim k_B \ln 2$), with an areal density set by a single UV cutoff length ℓ:

$$\eta \sim \frac{\ln 2}{\ell^2}. \tag{46}$$

Feeding this into the Jacobson relation $G = c^3/(4\hbar\eta)$ gives

$$G \sim \frac{c^3 \ell^2}{4\hbar \ln 2}. \tag{47}$$

Demanding the measured value of G then *fixes the cutoff*:

$$\ell^2 \sim \frac{4\hbar G \ln 2}{c^3} = 4 \ln 2\, \ell_P^2 \quad \Longrightarrow \quad \ell \sim \ell_P = \sqrt{\frac{\hbar G}{c^3}}. \tag{48}$$

This is exactly where the attempt fails. To get the right G the link density must be the Planck density, $\ell \sim \ell_P$ — but ℓ_P contains G by definition. We have not computed η from G-free QFT inputs; we have re-expressed G in terms of a cutoff that itself encodes G. The bond energy $E_{\text{bond}} = \hbar c/\ell$ at this cutoff is the Planck energy, which tells the same story. This is the identical circularity that defeats the Jacobson–Padmanabhan–Verlinde programs at the step of fixing the area-law coefficient: nothing in the local entanglement counting, as currently formulated, sets the cutoff without already knowing G. A genuine derivation needs an independent, G-free determination of either ℓ or η — for instance from a finite, regulator-independent entanglement entropy density of the relevant QFT — which we do not have.

9.4 Why the Bohr Magneton Is the Real Local Test

Although a local derivation of G is an open problem, the same mechanism applied at the electron Compton scale produces an exact, local, G-independent result: $\mu_B = e\hbar/(2m_e)$ (companion paper). This is the real evidence that the mechanism is on solid local ground; what's missing is the bridge from "the mechanism is locally real" to "the mechanism, applied to the cosmic ensemble, fixes G."

The honest position is therefore: the local seed is proven (Bohr magneton). The orchard's shape (Mach–Sciama at cosmic scale) is consistent with the same seed and with a specific predicted prefactor, but does not by itself constitute a derivation of G.

10 COMPUTATIONAL VERIFICATION

All derivations are implemented in Python scripts:

- `derive_G.py`—Main G derivation
- `derive_bond_energy.py`—$E_{\text{bond}} = \hbar c/r$ derivation
- `derive_3d_extension.py`—3+1D extension via Huerta & van der Velde
- `verify_symbolic.py`—Symbolic (`sympy`) proof of the $\langle 1/r \rangle$ integrals, the discrete 4/5 vs 16/15 prefactors, and the energy-balance chain (showing $\hbar$, m_p, κ cancel exactly)
- `gfree_inputs.py`—Evaluates the prefactor on purely G-free inputs (expansion-kinematics radii; CMB+BBN baryon census), quantifying the circularity
- `constants.py`—Physical constants and helper functions

All scripts run cleanly and produce consistent results. The symbolic check makes the core arithmetic exact rather than numerical: run `python3 verify_symbolic.py`.

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